MÓDULO 2: SEMÁNTICA

HyperNCD



$$\mathbf{v}_i = \lfloor x_i/\ell \rfloor,\ a_{\mathbf{v}} = 1$$

$$f_{\text{sem}}(x^i) = U_\theta(\mathcal{V})_{\mathbf{v}_i} \in \mathbb{R}^{96}$$

$$z_{i,k} = \frac{\langle f_{\text{sem}}(x^i), w_k \rangle}{\|f_{\text{sem}}(x^i)\|\,\|w_k\|}$$

$$Y = B\,\text{diag}(u)\,Q\,\text{diag}(v)$$

$$g^{(r)} = \left[\tfrac{\lambda_1-\lambda_2}{\lambda_1}, \tfrac{\lambda_2-\lambda_3}{\lambda_1}, \tfrac{\lambda_3}{\lambda_1}, c^{(r)}\right]$$

$$\hat{F}^{(r)} = \textstyle\sum_{j\in\mathcal{N}_M(r)} \tilde{S}_{rj} F^{(j)}$$

$$\mathcal{L} = \mathcal{L}_{\text{pt}} + \beta\,\mathcal{L}_{\text{reg}}$$

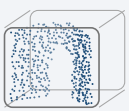
f_{sem} 96 por punto

nube de puntos en .LAS

coordenadas (x, y, z) de S

MÓDULO 1: PREPROCESAMIENTO

levantamiento TLS



$$x'_i = x_i - c$$

$$\bar{d}_i \le \mu + \lambda\,\sigma$$

$$\bar{x}_v = \tfrac{1}{|v|}\textstyle\sum_{i\in v} x_i$$

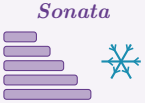
$$C = \tfrac{1}{q}\textstyle\sum_j (p_j - \bar{p})(p_j - \bar{p})^\top$$

$$S = [\,x, y, z, R, G, B, n_x, n_y, n_z\,]$$

S: 9 columnas

MÓDULO 3: ENCODER

Sonata



$$\mathbf{h}_i^{(0)} = \text{GELU}(\text{LN}(W_e \mathbf{x}_i + b_e))$$

$$\text{Atn}(\mathbf{H}_p) = \text{softmax}\big(\mathbf{H}_p W_Q (\mathbf{H}_p W_K)^\top/\sqrt{d}\big)\,\mathbf{H}_p W_V$$

$$\tilde{f}_j = [\mathbf{h}^{(3)}; \mathbf{h}^{(4)}; \mathbf{h}^{(5)}]$$

$$f_{\text{ssl}}(x^i) = \tilde{f}_{\text{inv}(i)} \in \mathbb{R}^{1088}$$

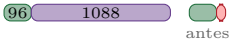
$$\hat{y}_i = \arg\max\,(W^\top f_{\text{ssl}}(x^i) + b)$$

f_{ssl} 1088 por punto

MÓDULO 4: FUSIÓN

HyperNCD

① Concatenación

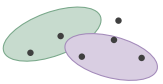


$$F_i = [\,f_{\text{sem}};\ f_{\text{ssl}}\,]$$

$$F^{(r)} = \tfrac{1}{|r|}\textstyle\sum_{i\in r} F_i$$

regiones de 0,75 m

② Afinidad entre regiones



$$S_s = \frac{\langle F^{(i)}, F^{(j)} \rangle}{\|F^{(i)}\|\,\|F^{(j)}\|}$$

$$S_g = \exp\left(-\|g^{(i)} - g^{(j)}\|^2/\tau\right)$$

$$S_b = \alpha S_g + (1-\alpha) S_s$$

S_b entre regiones

③ Hiperaristas

$$\hat{F}^{(r)} = \textstyle\sum_{j\in\mathcal{N}_M(r)} \tilde{S}_{rj} F^{(j)}$$

hipergrafo $\mathcal{H}$ $\hat{F}^{(r)}$

④ Prototipos y pseudo-etiquetas



$$P_k = \frac{\sum_i s_{i,k} F_i}{\sum_i s_{i,k}}$$

$$Q_{ik} = \text{softmax}(o_i/0{,}1)_k^{1/\epsilon_{\text{sk}}}$$

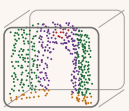
$$u \leftarrow \tfrac{1/B}{Qv},\ v \leftarrow \left(\tfrac{1/|\mathcal{C}_n|}{Q^\top u}\right)^{\frac{\gamma}{\gamma+\epsilon_{\text{sk}}}}$$

$$Y = B\,\text{diag}(u)\,Q\,\text{diag}(v)$$

etiquetas $\hat{y}_i$

MÓDULO 5: SALIDA

evaluación y plataforma



$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c}$$

$$\text{mIoU}(\mathcal{S}) = \tfrac{1}{|\mathcal{S}|}\textstyle\sum_{c\in\mathcal{S}} \text{IoU}_c$$

c^* σ^*

ARI NMI

$$\Delta\text{mIoU}^{\text{novel}} = \text{mIoU}^{\text{novel}}_{\text{hib}} - \text{mIoU}^{\text{novel}}_{\text{base}}$$

$$G = \frac{\Delta\text{mIoU}^{\text{novel}}}{\text{mIoU}^{\text{known}}_{\text{base}} - \text{mIoU}^{\text{novel}}_{\text{base}}}$$

→ conexión implementada